# Gary Kocsis

**Protocol Engineer | Smart Contract Engineer**

Available for NYC hybrid roles • 610-554-9722 • gkocsis99@gmail.com

Portfolio • LinkedIn • GitHub • Standby • RangeGuard

## PROFESSIONAL SUMMARY

Protocol Engineer and Smart Contract Engineer focused on secure DeFi protocols. Applies decades of experience building financial systems to smart contract architecture, defining specifications and invariants before implementation and verifying them through testing.

## OPEN SOURCE PROTOCOL ENGINEERING

### Standby - Uniswap v4 Protocol | ETHOnline 2026, ETHGlobal

- Designed and implemented Standby, a Uniswap v4 protocol providing bounded future execution capacity from shared AMM liquidity without dedicated reserves or pre-positioned liquidity.
- Defined the protocol architecture, state transitions, and trust boundaries from an explicit economic agreement. Enforced the backing invariant that supporting AMM capacity must cover outstanding obligations.
- Verified the backing invariant and exercise lifecycle through fuzz and stateful invariant testing with independent reference calculations; deployed to Base Sepolia and executed the canonical lifecycle through the official Uniswap stack.

### RangeGuard - Uniswap v4 Protocol | Atrium Uniswap Hook Incubator

- Designed and implemented a Uniswap v4 hook that funds impermanent loss coverage with a portion of each swap fee and calculates accrual using the Actual/365 Fixed convention. Verified coverage accounting with Foundry unit, fuzz, and invariant tests.

## PROFESSIONAL EXPERIENCE

### Macro | Instruction Resident · Contributor | Remote | March–November 2022

- Reviewed student Solidity smart-contract projects for security vulnerabilities and code-quality issues, documented potential findings in shadow-spec documents, and provided actionable feedback to strengthen contract design. Selected security reviews.

**JPMorgan Chase - Vice President** | 2009–Present

- Managed onshore and offshore consultants developing conversion programs to migrate more than one million Chase brokerage accounts from Fidelity's National Financial Services (NFS) platform to JPMorgan Chase. Wrote SQL to verify mapped field values side by side.
- Designed and built the Required Minimum Distribution (RMD) questionnaire application using DB2 and COBOL stored procedures. Coordinated conversion job execution for the migration of 815,000 retirement accounts.
- Developed the electronic prospectus delivery process for a campaign that increased paperless adoption from 1% to 11% across more than 800,000 customers.

**Bear Stearns - Associate Director** | 1998–2008

- Architected migration to FIX XP2, enabling global 24x7 trading while reducing client downtime from approximately three hours to under five minutes.
- Designed cross-domain routing and direct exchange connectivity for institutional FIX clients.
- Designed a FIX trade capture platform that processed approximately 90% of the firm's trading volume for regulatory reporting and enterprise audit trails.

## TECHNICAL SKILLS

**Protocol Engineering:** Solidity • Foundry • Uniswap v4 • Protocol Architecture • State Machines • Invariants • Fuzz/Invariant Testing • Token Engineering

**Engineering:** TypeScript • Java • Python • SQL • AWS • Kafka • Kubernetes • Terraform

## EDUCATION

MBA, DeSales University

B.S. Applied Mathematics-Economics, Brown University

## PROFESSIONAL DEVELOPMENT

**Macro Fellowship Engineering Program** | January–February 2022

Completed a six-week Ethereum/Solidity bootcamp focused on security and auditing, building an AMM, a DAO with ECDSA voting, and token fundraising contracts.

**Token Engineering Academy** | Token Engineering Fundamentals | April–June 2024

**Stanford Center for Professional Development** | Certificate in Advanced Computer Security | 2018

**Certifications:** AWS Solutions Architect Professional • AWS DevOps Engineer Professional • Terraform Associate